CSE STUDY HUB

Data Structures & Algorithms

UNIT 6 · Data structures in AI

What this unit is about

This unit builds understanding of **Data structures in AI**. These notes are designed for quick learning and exam revision: learn the meaning, understand the working, practise an example, and finish with comparisons and key terms.

Syllabus coverage checklist

1. Heuristic search and A* algorithm: g(n), h(n) and f(n)=g(n)+h(n)

2. Knowledge graphs: entities, relationships, triples and graph queries

3. Graph Neural Networks (GNNs): nodes, edges, message passing and applications

4. How trees, heaps, hash maps and graphs support AI systems

5. AI applications: recommendation, route planning, semantic search and knowledge representation

6. Limits of heuristic and graph-based methods; explainability and scalability

How to study: Read each topic once for meaning, underline definitions and formulas, reproduce diagrams/steps without looking, then answer the revision questions on Page 5.

Exam answer pattern

For a 5-mark answer: (1) definition, (2) labelled diagram or formula, (3) working/components in steps, (4) example/application, and (5) one advantage, limitation or comparison.

UNIT 6 NOTES · PART 1

Heuristic search and A* algorithm: g(n), h(n) and f(n)=g(n)+h(n)

Present Heuristic search and A* algorithm: g(n), h(n) and f(n)=g(n)+h(n) as an ordered workflow. State the preconditions, list the major steps, explain what each step produces, and mention how errors are detected or handled. Use a small worked example or flowchart in the exam.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Knowledge graphs: entities, relationships, triples and graph queries

A graph contains vertices and edges. An adjacency matrix uses $O(V^2)$ space; an adjacency list uses $O(V+E)$. BFS explores level by level with a queue; DFS explores deeply using recursion or a stack.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Data structures in AI, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 6 NOTES · PART 2

Graph Neural Networks (GNNs): nodes, edges, message passing and applications

A graph contains vertices and edges. An adjacency matrix uses $O(V^2)$ space; an adjacency list uses $O(V+E)$. BFS explores level by level with a queue; DFS explores deeply using recursion or a stack.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

How trees, heaps, hash maps and graphs support AI systems

A tree is a hierarchical, connected acyclic structure. Learn root, leaf, parent, child, depth and height. Traversals visit nodes in a defined order; BST ordering enables efficient search when balanced.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Data structures in AI, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 6 NOTES · PART 3

AI applications: recommendation, route planning, semantic search and knowledge representation

For AI applications: recommendation, route planning, semantic search and knowledge representation, connect the concept to a practical scenario. Identify the input or need, the technology/process used, the output or benefit, and one limitation. Strong examples explain not only what is used but why it is suitable.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Limits of heuristic and graph-based methods; explainability and scalability

A graph contains vertices and edges. An adjacency matrix uses $O(V^2)$ space; an adjacency list uses $O(V+E)$. BFS explores level by level with a queue; DFS explores deeply using recursion or a stack.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Data structures in AI, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

QUICK REVISION & EXAM PRACTICE

Key takeaways

• **Heuristic search and A* algorithm: g(n), h(n) and f(n)=g(n)+h(n)** — know the definition, core working and one example.

• **Knowledge graphs: entities, relationships, triples and graph queries** — know the definition, core working and one example.

• **Graph Neural Networks (GNNs): nodes, edges, message passing and applications** — know the definition, core working and one example.

• **How trees, heaps, hash maps and graphs support AI systems** — know the definition, core working and one example.

• **AI applications: recommendation, route planning, semantic search and knowledge representation** — know the definition, core working and one example.

• **Limits of heuristic and graph-based methods; explainability and scalability** — know the definition, core working and one example.

Worked-answer framework

Scenario: A question asks you to explain or apply a concept from Data structures in AI. **Step 1:** identify the relevant term and define it. **Step 2:** write the formula, components or ordered process. **Step 3:** apply it to the given data/scenario. **Step 4:** state the result and justify it. **Step 5:** mention one limitation or precaution.

Self-test questions

Q1. Define Heuristic search and A* algorithm: g(n), h(n) and f(n)=g(n)+h(n) and explain its importance.

Q2. Explain the working/steps involved in Knowledge graphs: entities, relationships, triples and graph queries.

Q3. Differentiate between two related concepts from Data structures in AI.

Q4. Give one practical application of Limits of heuristic and graph-based methods; explainability and scalability and discuss one limitation.

Q5. Write a structured 5-mark answer covering the main ideas of Data structures in AI.

Final checklist: I can define every topic · explain the process without notes · reproduce formulas/diagrams · solve one example · compare related concepts.